Jeremy Mattathias Mboe

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EDUCATION

2024-present **Bachelor of Artificial Intelligence Engineering, GPA: 3.86** (lowest grade: 0.00, highest grade: 4.00)
Institut Teknologi Sepuluh Nopember (ITS), Surabaya, Indonesia. Expected graduation: 08/2028
Relevant courses: Probability and Statistics, Machine Learning, Deep Learning, Computational Intelligence, Data Mining, NLP, Digital Image Processing

RESEARCH EXPERIENCE

2026-present **Laboratory Assistant**
Intelligent Computing and Vision Laboratory, ITS. Supervisor: Dini Adni Navastara
3D segmentation of kidney tumors and cysts in a five-student Kerja Praktik research team, in collaboration with clinicians from Rumah Sakit Universitas Airlangga. Prepare practicum modules for Machine Learning and Data Mining courses.

2026-present **Research Assistant, AI Division**
Diagnostic and Prognostic Laboratory, ITS.
3D CT reconstruction from triplanar chest X-rays (AP, PA, lateral) for tuberculosis screening, using physics-informed neural networks and neural attenuation fields that model the Beer-Lambert law for sparse-view tomography.

2026-present **Deep Learning Researcher, SwaraTTS**
Avalon AI x RISTEK Universitas Indonesia
Low-latency neural speech synthesis with score-based diffusion and latent flow matching. Compared continuous normalizing flows with diffusion-based sampling for efficiency.

2025-present **Machine Learning Developer, TB Cough Detection and Diagnostic System**
Diagnostic and Prognostic Laboratory, ITS. Supervisor: Dhany Arifianto
Energy-based cough segmentation, MFCC features and a two-layer LSTM on 2,244 cough recordings. About 90% accuracy and F1-score 0.89 on 449 held-out recordings.

2026 **Software Developer Intern**
Pusat Studi Kecerdasan Artifisial dan Teknologi Digital, ITS
Built an EEG processing pipeline for 1,809 clinical EDF recordings, from task segmentation to frequency-band features. Developed a binary encoding of EEG signals to classify left- vs. right-hand movement intention, and compared machine learning models with an EEGNet baseline.

SELECTED RESEARCH PROJECTS

2026-present **Foundational Neuromorphic Vision Research** (AI Research Assistant, Avalon AI Corporation)
Image classification with spiking neural networks using Izhikevich neuron dynamics and temporal spike encoding. Built the full training pipeline from scratch in PyTorch without SNN frameworks: custom surrogate gradient functions to train through non-differentiable spikes, temporal encoders and spike-based layers. Using this custom pipeline, the model reached 76.6% accuracy on the N-MNIST dataset.

2026 **AVID-MM: Audio-Visual Identity Disentanglement** (Lead Deep Learning Researcher)
Cross-modal speaker verification that stays robust when audio or video is missing.

ECAPA-TDNN voice embeddings with pretrained face backbones raised accuracy from 59% to over 93% on the POLY_1 challenge setup.

2025 **Oil Palm Tree Health Detection from Drone Images** (Deep Learning Researcher) YOLOv8s detection of 11,044 annotated trees (precision 96.7%, recall 97.2%) and ResNet18 health classification on RGB-estimated NDVI crops (92.4% accuracy). Deployed with Gradio on Hugging Face Spaces.

AWARDS AND SCHOLARSHIPS

2025 1st Place and Best Scientific Paper, National Electrolympic, Universitas Pembangunan Nasional Veteran Jakarta (TB cough detection system)

PUBLICATIONS

Poster presentations

Jeremy Mattathias Mboe et al.Mortality Prediction in Tuberculous Meningitis Using XGBoost and a SHAP-Derived Nomogram. *ASEAN4TB Conference*, 2026 (e-poster). Interpretable XGBoost model (AUROC 0.825) and an 8-feature SHAP-derived logistic regression nomogram (AUROC 0.815) for early 3-month mortality stratification in tuberculous meningitis, evaluated with Decision Curve Analysis.

CONFERENCES, WORKSHOPS, COMPETITIONS, OTHER RESEARCH ACTIVITIES

2026 ASEAN4TB Conference (e-poster presented as primary author and presenter)
2026 SATRIA DATA national data science competition (participant)
2025 National Electrolympic UPNVJ (paper presented to a panel of academic experts)
2025 GEMASTIK national competition, Data Mining division (participant)

TEACHING EXPERIENCE

2026-present Teaching Assistant for BSc courses "Probability and Statistics", "Machine Learning" and "Data Mining"
Department of Informatics, ITS

RELEVANT EXTRACURRICULAR EXPERIENCE

2025-present Computer Vision Specialist, Avalon AI Group (intensive training in modern computer vision architectures)
2025-present Schematics ITS: Data Management staff (2025) and Expert Staff, Data Science Sub-Division (2026), national data science event
2025-2026 Backend AI Engineer, REXTRA: REST API infrastructure for LLM inference
2024 Programming Division Intern, IRIS ITS robotics team: C++, OpenCV and ROS for robot movement control

COMPUTING SKILLS

Python	advanced (PyTorch, NumPy, Pandas, scikit-learn, Hugging Face)
ML frameworks	intermediate (TensorFlow, JAX)
C++, SQL	intermediate (OpenCV, ROS, PostgreSQL)
Tools	intermediate (Git, Linux, Docker, Jupyter)
Web	basic (JavaScript, Django, FastAPI)

LANGUAGE SKILLS

Indonesian	native
English	fluent (professional working proficiency)

CERTIFICATES

2025 - 2028 Machine Learning and Deep Learning Developer certification (TensorFlow: model training, evaluation and deployment)